COMSATS University, Islamabad

Department of Computer Science

Lab Mid (Version 1) - SP26

Instructor: Ms. Sajida Kalsoom

All questions are mapped to CLO4

Imagine a small but popular online electronics store run by **Hamza**. Many customers visit the store to purchase gadgets from different categories. Hamza wants a system to manage the gadgets available in his store and handle customer orders efficiently. The system should track product details such as name, brand, price, and category, while ensuring that important information is validated and protected from invalid values. At the same time, customers should be able to easily view the details of each gadget.

Each product, such as a laptop or smartphone, has specific details including a **name, brand, price, and category**. Hamza wants these details to remain protected so that they can only be accessed or updated through controlled methods, and the system must reject invalid values, such as a zero or negative price, when they are set.

Sometimes Hamza sells gadgets together as a single order, such as a "**Work From Home Kit**," which includes a laptop along with one or more smartphones. Therefore, the system should be capable of **placing an order for a single product as well as for multiple products using the same operation.**

Hamza also organizes gadgets into **broader categories**. For example, all products share common properties such as **name, brand, and price**, but certain types of gadgets have **additional features**. A laptop might include a processor type and RAM size, while a smartphone might include a camera resolution and battery capacity. **A general product should never exist on its own in the store.** The system should allow staff to **display details of any product in a clear format** and also **compare two products** to determine if they are the same.

When a customer places an order, the store sends the **complete order object** to the **billing system**, which processes the information, **calculates the total bill**, and generates the receipt for the customer.

To implement this solution, Hamza hires a programmer who develops a system that models **products and their relationships**, ensuring the design is **flexible enough** to support new types of gadgets in the future, such as tablets or smartwatches, while keeping important data secure and allowing easy comparison and display of product details.

COMSATS University, Islamabad

Department of Computer Science

Lab Mid (Version 2) - SP26

Instructor: Ms. Sajida Kalsoom

All questions are mapped to CLO4

Imagine a small but popular online bookstore run by **Hamdan**. Many customers visit the store to purchase books from different categories. Hamdan wants a system to manage the books available in his store and handle customer orders efficiently. The system should track book details such as title, price, author, and genre, while ensuring that sensitive information can only be modified by authorized staff. At the same time, customers should be able to easily view basic information like the book title and price.

Each **book**, such as a novel or textbook, has specific details including a **title, author name, price, and genre**. Hamdan wants these details to remain protected so that **only store administrators can update or modify them**, while customers can only access general information when browsing the store.

Sometimes Hamdan offers special bundles, such as "**Hamdan's Exclusive Study Pack for Students**," which includes a textbook along with a notebook and a bookmark. Therefore, the system should be capable of **handling such combinations smoothly.**

Hamdan also organizes books into **broader categories**. For example, all books share common properties such as **title and price**, but certain types of books have **additional features**. A textbook might include an edition number, while a novel might include a series name. The system should allow staff to **display details of any book in a clear format** and also **compare two books** to determine if they are the same.

When a customer places an order, the store staff sends the complete order object to the **billing system**, which processes the information and **generates the receipt** for the customer.

To implement this solution, Hamdan hires a programmer who develops a system that models **books and their relationships**, ensuring the design is **flexible enough** to support new types of items in the future, such as magazines or e-books, while keeping important data secure and allowing easy comparison and display of book details.